

## Exp no:9

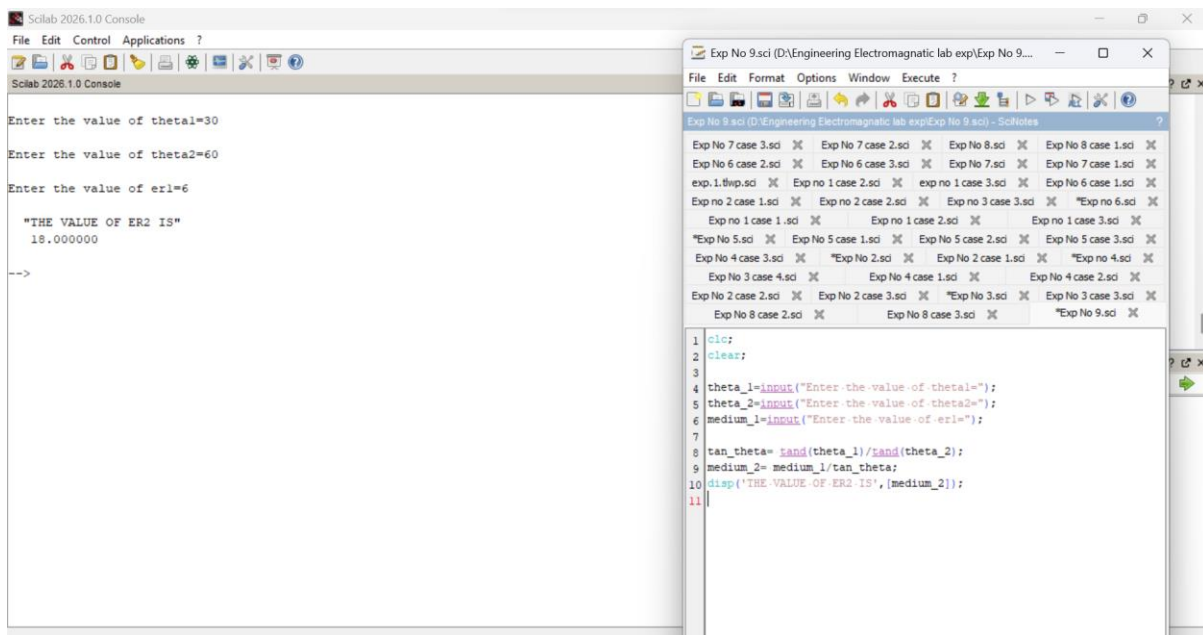
**Implement Dielectric free space boundary conditions in Electric Field and find the relative permittivity of the different medium.**

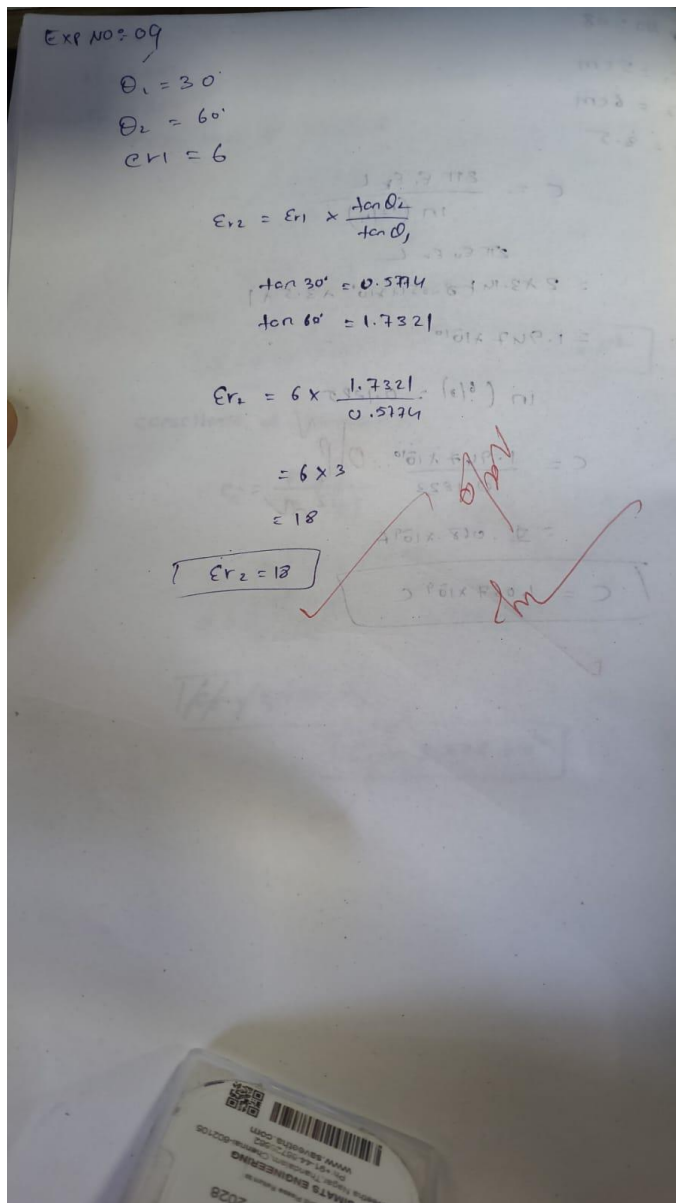
### Aim:

To write program for Dielectric - free space boundary condition in Electric field using SCILAB.

### Software required:

- SCILAB version 6.0.1





**Case 1:** In this case, the dielectric medium has a relative permittivity ( $\epsilon_r$ ) of 2, meaning the dielectric reduces the electric field by half compared to the free space. This is typical for materials like mica or certain types of plastics.

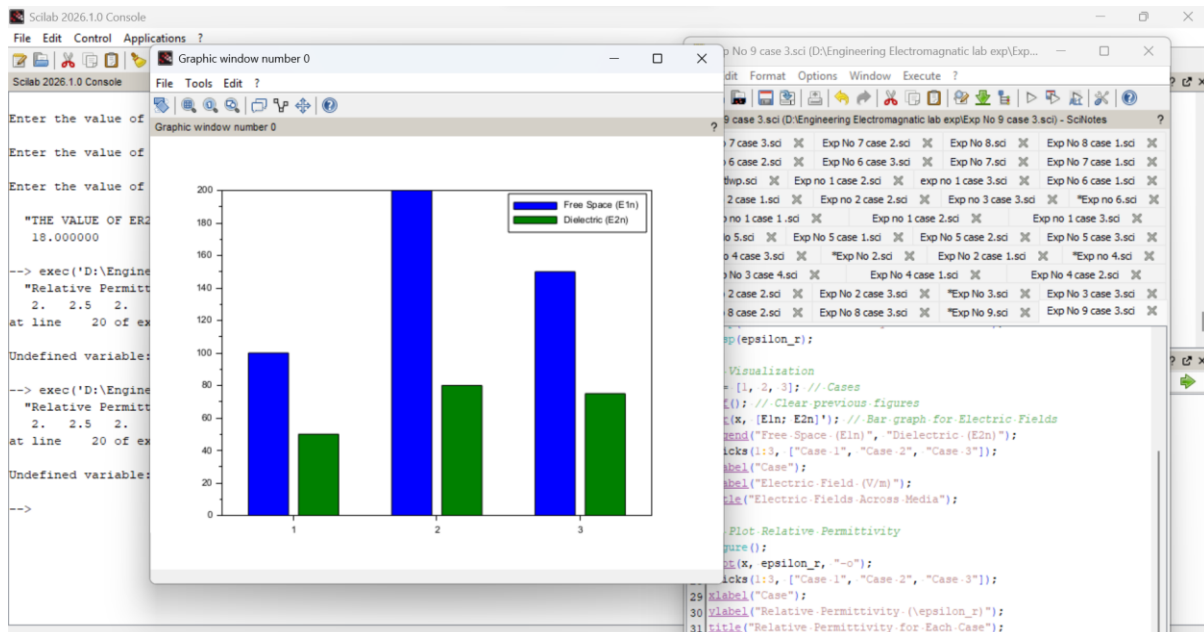
- **Free Space Electric Field ( $E_1$ ):** 100 V/m
- **Dielectric Electric Field ( $E_2$ ):** 50 V/m

**Case 2:** Here, the dielectric medium has a higher relative permittivity ( $\epsilon_r=2.5$ ) compared to Case 1. This indicates a stronger polarization effect in the dielectric, as seen in materials like glass or water.

- **Parameters:**
  - $E_{n1}=200$  V/m (Electric field in free space).
  - $E_{2n}=80$  V/m (Electric field in the dielectric medium).

**Case 3:** The dielectric medium here also has a relative permittivity of 2. The behaviour of the material suggests it could be another low-polarization dielectric material, with a consistent reduction of the electric field by half.

- **Free Space Electric Field ( $E_{1n}$ ):** 150 V/m
- **Dielectric Electric Field ( $E_{2n}$ ):** 75 V/m



## Result :

Thus the program was verified for Dielectric - free space boundary condition in Electric field using SCILAB was successfully.